

Apache, PHP, FTP and Node.js Assignment

Question 1: Install Apache web server and PHP on your Linux machine, then create an index.php file that displays 'Welcome to My Foodie Blog'.

Answer:

Install Apache and PHP:

Ubuntu/Debian:

```
sudo apt update
```

```
sudo apt install apache2 php libapache2-mod-php -y
```

Start Apache:

```
sudo systemctl start apache2
```

```
sudo systemctl enable apache2
```

Create index.php:

```
sudo nano /var/www/html/index.php
```

Add:

```
<?php
```

```
echo "Welcome to My Foodie Blog";
```

```
?>
```

Restart Apache:

```
sudo systemctl restart apache2
```

Test:

```
http://SERVER_IP/index.php
```

Question 2: Configure Apache to serve a custom 404 error page.

Answer:

Create Error Page:

```
sudo nano /var/www/html/404.html
```

Content:

Oops! This page is as empty as your cart.

Edit Apache Configuration:

```
sudo nano /etc/apache2/sites-available/000-default.conf
```

Add:

```
ErrorDocument 404 /404.html
```

Restart Apache:

```
sudo systemctl restart apache2
```

Test:

Open a non-existing URL.

Question 3: Set up an FTP server and upload a sample image file.

Answer:

Install VSFTPD:

```
sudo apt install vsftpd -y
```

Start Service:

```
sudo systemctl enable vsftpd  
sudo systemctl start vsftpd
```

Upload File Using FTP Client (FileZilla):

Host: SERVER_IP
Username: yourusername
Password: yourpassword
Port: 21

Upload image file to:

/var/www/html/

Verify:

http://SERVER_IP/image.jpg

Question 4: Configure a basic Node.js service on port 3000 that returns JSON data.

Answer:

Install Node.js:

```
sudo apt install nodejs npm -y
```

Create app.js:

```
const http = require('http');  
http.createServer((req,res)=>{  
  res.writeHead(200,{ 'Content-Type': 'application/json' });  
  res.end(JSON.stringify({playlist:['Song1','Song2','Song3']}));  
}).listen(3000);
```

Run Service:

```
node app.js
```

Test PHP Site:

http://SERVER_IP/index.php

Test Node.js API:

http://SERVER_IP:3000

Expected JSON:

```
{"playlist":["Song1","Song2","Song3"]}
```

Question 5: Restrict access to /var/www/html/secret so that only localhost can access it.

Answer:

Create Directory:

```
sudo mkdir /var/www/html/secret
```

Edit Apache Configuration:

```
sudo nano /etc/apache2/apache2.conf
```

Add:

```
<Directory "/var/www/html/secret">
```

```
  Require ip 127.0.0.1
```

```
</Directory>
```

Restart Apache:

```
sudo systemctl restart apache2
```

Test:

Localhost can access the directory.

Other systems receive Access Denied (403 Forbidden).